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# Automation of Data Flow Management Based on Artificial Intelligence in Systems with an Internal Distribution Mechanism

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***Abstract***

This research work is devoted to the development of optimal management approaches to data flows based on artificial intelligence algorithms in systems with an internal distribution mechanism. The study first discusses the internal distribution mechanism, its capabilities, and the design necessary to achieve efficiency. During the research, an algorithm for managing data flows in the internal distribution mechanism is developed. In addition, the use of artificial intelligence algorithms that are machine learning algorithms and multilayer neural networks is considered to implement the steps of this algorithm. This thread looks at identifying the most effective machine learning algorithms and multilayer neural network architectures for research. The research also discusses the methods of distribution of data in the internal distribution mechanism. It also provides information about the methodology of the research process and the computers used in the research. The results obtained in the research

are described and analyzed. The main conclusions based on the results obtained at the end of the research are presented in detail.

***Keywords*:** Data flow management algorithm, internal distribution mechanism, machine learning, multilayer perceptron, intelligent automation systems, clustering, distribution algorithms

## 4.1 Introduction

### 4.1.1 The Importance of the Research and the Setting of the Problem

Today, in the developed countries of the world, all data exchange is carried out digitally. The presence of information in digital form creates many opportunities for humanity [1, 2]. Digital information makes it possible to prevent red tape and organize quick and efficient work in all areas. In addition, digital information creates an opportunity to ensure transparency and openness in state organizations—education, medicine, and socio-economic spheres [3–5]. For this reason, the process of digitization is being rapidly continued in many developing countries as well as developed countries in the world. The process of data digitization can also be seen in production and service enterprises operating in the private sector. Most of the digitized data in these fields is structured data with a specific structure [6, 7]. It is known that storing, processing, and analyzing structured data requires less time and cost than unstructured data [8, 9]. But as a result of the use of these information systems by thousands and millions of people, structured data streams are created that are difficult to manage using traditional methods and tools [10]. In this process, as a result of the constant creation of new data and their addition to the database, it creates large volumes of structured data streams in information systems.

Sources of large volumes of structured data streams are not only information systems of organizations and enterprises but also unstructured or semi-structured data flows that can be modified for analysis [11, 12]. It is known that the process of direct analysis of unstructured or semi-structured data and the identification of hidden patterns in them is difficult or in some cases impossible. In such cases, this information is transferred to a certain structure and then analyzed. Even in this case, the analyzed data, when transferred to a certain structure, has a large volume, so it creates data streams.

In general, most of today's information systems are forced to work with large data streams [13]. However, existing methods and approaches are unable to manage data flows known as big data [14–16]. Therefore, the development of methods and algorithms that solve the problems of large data flows is one of the current argent research topics. Today, many research works on this current topic can be found [17–21]. However, although most of them have productivity in terms of time, they require a large amount of economic costs. Methods and approaches that are both economically and time-efficient are currently rare [22–26]. One such ap-

proach is the approach based on the internal distribution mechanism. Although this approach, as mentioned above, is efficient in terms of time and economy, it also has some complications [27–30]. In other words, in order for it to be effective in terms of time, it is required to perform the tasks performed in it perfectly. However, the changeable nature of data limits human capabilities. In order to overcome this problem, this research paper considers the development of an artificial intelligence-based algorithm for managing data flows in the internal distribution mechanism. During the study, the essence of distributed and internally distributed computing approaches will be covered. It provides detailed information about the capabilities of the internal distribution mechanism and the design necessary to achieve efficiency. In the next steps of the research, an algorithm for managing data flows in the internal distribution mechanism will be developed. During the study, the use of artificial intelligence algorithms to implement the steps of the data flow management algorithm is considered. This thread looks at identifying the most effective machine learning algorithms and multilayer neural network architectures for research. The research also discusses the methods of distribution of data in the internal distribution mechanism. At the end of the research, the results obtained will be described and analyzed, and the main conclusions based on the obtained results will be presented in detail.

### 4.1.2 Distributed Computing Systems

It is difficult to process and analyze large data streams in real time based on traditional methods and approaches [31–33]. In this case, although the structure of the tables in the database meets all normalization requirements, as the volume of data, i.e., the number of records in the tables, increases, the time required to perform queries on this data also increases. The main reason for this is that it is not possible to directly refer to the information expected as a result of a query in the database. In other words, the query space covers all the records in the table. That is, an increase in the number of records by one causes the next query space to increase by one unit, and an increase in the number of records to a million causes the query space to increase by a million units. Therefore, the query execution time increases in proportion to the number of records.

In this case, one of the main solutions for processing and analyzing large data streams is to store the data streams in several small pieces in distributed computing systems [26, 34–36]. Data processing and analysis using a distributed computing system is based on the approach of storing data horizontally or vertically on several servers [37–41]. In this, each server has its own memory and operating system, and when a request is made by the clients, the servers gain efficiency by dividing the requests or working together [42–44]. An overview of the architecture of storing and processing data streams in a distributed computing system is presented in Figure 4.1.

A distributed computing approach can achieve high-time performance on large data streams. However, since several computing machines are required to create this computing system, the implementation costs of this

system are high. In some cases, the costs of using a distributed computing system for small enterprises and organizations exceed the income received as a result of data processing and analysis, or the costs incurred cannot be covered in a short time [45, 46]. Therefore, this approach is not considered a satisfactory solution for representatives of small businesses, which are the basis of the economy. Therefore, the development of both time- and cost-effective approaches to the storage, analysis, and processing of large data streams is one of the current research topics.

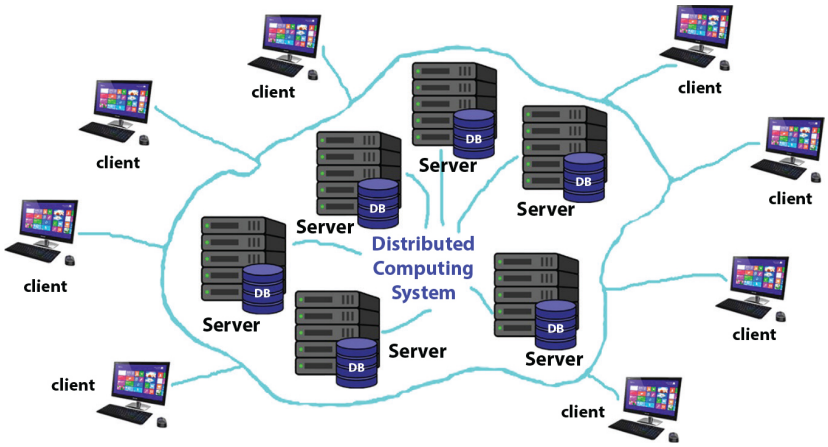



**Figure 4.1** The general architecture of a distributed computing system.

### 4.1.3 Internal Distribution Mechanism

The internal distribution mechanism is based on a distributed computing approach. However, using a single server instead of multiple servers is sufficient in this internal distribution approach [27]. In distributed computing systems, data and queries are distributed over several servers, while in the internal distribution approach, they are distributed across tables with horizontal distribution within the database on a single server. When a client makes queries, the distribution model forwards the queries to the appropriate table in the database. The distribution rules used to divide the data into tables are used to select the appropriate table. During the execution of the request, the search is not performed among all the data on the server, only this process is performed in a certain table. The fact that the query is not performed on all data has a positive effect on the data processing time. That is, if the number of distributions is N, the query time can be reduced to a maximum of N times [27]. The architecture of data flow storage and processing based on the internal distribution mechanism is presented in Figure 4.2.

As shown in Figure 4.2 below, requests from clients are first sent to the distribution module. The distribution module processes the submitted request and forwards the request to the required horizontally distributed table based on the existing distribution rules. The distribution module plays an important role in ensuring reliability and accuracy in this process. Therefore, it is necessary to properly organize the functional task of the distribution module. A small error in the distribution module will negatively affect the reliability of the query result and in some cases may cause the system to crash. The distribution module task can be organized in two ways [27]. In the first method, it is done using standard program-

ming methods. It is advisable to use this method when the size and characteristics of the stored data are known in advance. The second method is based on artificial intelligence algorithms. In this method, the optimal number of distributions and the distribution module's request distribution rules are developed on the basis of artificial intelligence algorithms, and allocation processes are carried out. This method is more efficient than the first method and allows to use the proposed approach even in cases where the volume of data and its characteristics are not known in advance. This method can also be seen in the functional scheme presented in Figure 4.3.

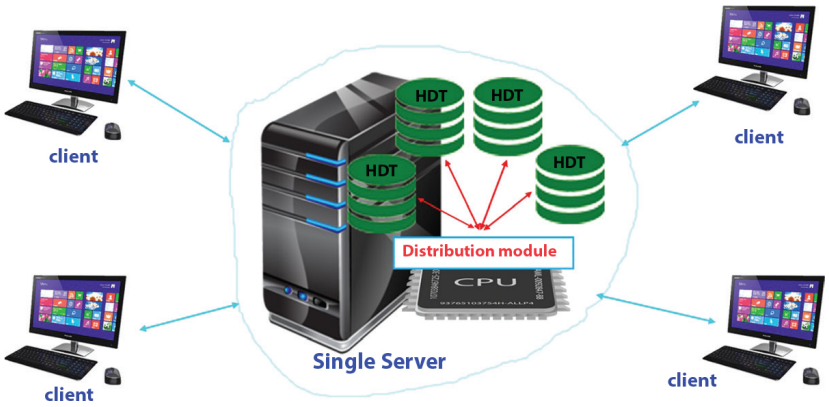



Figure 4.2 The general architecture of a distributed computing system.

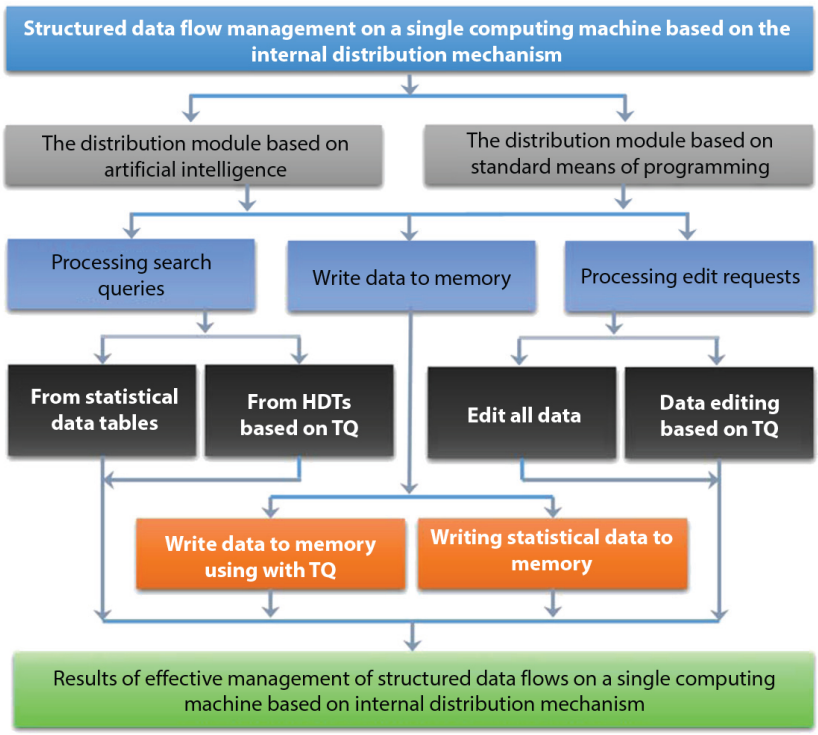



Figure 4.3 Functional scheme of data management on a single server based on internal distribution mechanism. TQ is a rule of queries and data distribution into horizontally distributed tables (HDTs).

As mentioned above, the internal distribution mechanism is not only economical but also time-efficient in storing and processing data streams. This can be clearly seen when comparing the time performance of this approach with the time performance of multiple computing systems without shared memory and the time performance of a single computing system in the traditional approach. If the request processing time of a dis-

tributed computing system consisting of several computing systems without shared memory is taken as t, it is found as follows [23]:

$$t = t_d + t_{I/O} + t_{CPU} + t_c, \tag{4.1}$$

where $t_d$ is the time required to distribute data or tasks to distributed computing systems, which is directly proportional to the number of computing machines in the system ($N$). That is

$$t_d = t_1 \cdot N, \tag{4.2}$$

where $t_1$ is the average time it takes to distribute data or tasks to one computing system.

$t_{I/O}$ is the time required for data input and output;

$t_{CPU}$ is the time required to complete tasks, which is directly proportional to the number of tasks. If the given task is to process data in the database, the total number of records in the tables in the database ($n(R_i)$) is taken as the number of task sets, and $t_{CPU}$ is expressed as follows:

$$t_{CPU} = max(n(R_1) \cdot t_2,\ n(R_2) \cdot t_2, n(R_3) \cdot t_2, \ldots, n(R_k) \cdot t_2), \tag{4.3}$$

where $R_i = \sigma_F(R)$ is a set of parts of complete database ($R$) divided by some condition ($F$) and stored on different computing machines in the system, $i \in [1..k]$, and $t_2$ is the time taken by the processor to execute a single instruction. If the computing machines in the distributed computing system are heterogeneous, that is, if the performance indicators of the processors are different, then $t_2$ will also be different;

$t_c$ is the time required to establish communication between computer systems and transfer data from one computer to another, and it is defined as follows:

$$t_c = t_3 + V \cdot t_4, \tag{4.4}$$

where $t_3$ is the time to establish a connection between computing systems, $V$ is the volume of data being transferred, and $t_4$ is the time required for data transfer per volume unit.

Based on the above, formula (4.1) is expressed as follows:

$$t = t_1 \cdot N + t_{I/O} + t_3 + V \cdot t_4 + max(n(R_1) \cdot t_2, n(R_2) \cdot t_2, n(R_3) \cdot t_2, \ldots, n(R_k) \cdot t_2) \tag{4.5}$$

Formula (4.5) is the request processing time of distributed computing systems consisting of several computing systems that do not share common memory. It can be seen from this formula that when determining the data processing time of the proposed internal distribution approach, $t_c$ is not taken into account because all processes are performed on a single computing machine. This means that the time for processing requests based on the internal distribution mechanism is at least $t_c$ less than in distributed computing systems. In other words, it is efficient to perform queries on the data stream based on the internal distribution mechanism compared to the distributed computing systems.

When calculating the time costs of the internal distribution mechanism, the number of horizontally distributed tables in the database of single

computing system ($N'$) is taken as $N$, and average time $t_1'$ is taken to distribute requests to one of the horizontally distributed tables as $t_1$. Instead of $R_i$, $DT_i = \sigma_S(DT)$ in the database of the query ($DT$) table, based on some ($S$) condition, partition sets distributed in a horizontally distributed table are obtained, $i \in [1..k]$. And $t_2'$ is invariant because it is in a single calculation system.

Taking into account the above, the request processing time ($t'$) on a single server based on the internal distribution mechanism is determined using the following formula (4.6):

$$t' = t_1' \cdot N' + t_{I/O} + max(n(DT_1) \cdot t_2', n(DT_2) \cdot t_2', \ldots n(DT_k) \cdot t_2') \quad (4.6)$$

Based on the traditional approach, when determining the time cost of processing requests on a single computing machine, the communication time cost and the request allocation time cost are equal to 0 (zero). Therefore, when calculating its time costs, $t_c$ and $t_d$ are not taken into account. Also, $t_{CPU}$ is equal to $n(DT) \cdot t'_2$. In general, time costs ($t''$) of data processing in traditional methods are calculated as follows:

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$$t'' = t_{I/O} + n(DT) \cdot t_2' \quad (4.7)$$

Due to the absence of $t_d$ in this formula (4.7), it seems that the traditional methods are more efficient than the internal distribution mechanism. However, since $DT = D_1 \cup DT_2 \cup DT_3 \cup \ldots \cup DT_k$, $n(DT)$ is several times larger than $max(DT_1 \cup DT_2 \cup DT_3 \cup \ldots \cup DT_k)$. It shows that the storage and processing of structured data flows on a single server based on the internal distribution approach can achieve several times more efficiency in terms of time cost compared to traditional methods.

The internal distribution approach can provide positive efficiency not only in terms of time or economy but also according to several indicators. It can also be seen from Table 4.1 [23].

It can be seen from Table 4.1 that the approach of storing and processing data streams on a single server based on the mechanism of internal distribution is better than the approach of distributed computing systems with several servers in terms of cost, security, controllability, integrity, and network reliability and is considered more effective. The availability indicator of this approach depends on the constant availability of electric current and protection from external influences, and this problem can be solved practically today. The main problem in the approach is the limitation of the amount of data to be stored and processed, that is, the problem of scalability. This is related to the memory resource of a single server.

It was mentioned above that an effective approach is to store and process data streams using an internal distribution mechanism. However, it should be noted that in the internal distribution approach, only the correct design of the distribution process will lead to the expected result.

Here, the right design of the distribution process means solving the three problems presented in Figure 4.4:

Issues that need to be solved for the design of the distribution module in the internal distribution mechanism [23, 27]:

1. Determining the number of HDT in the database in proportion to the volume of data. Because increasing the number of horizontally distributed tables does not always minimize query processing time, the time required for the task performed by the distribution module also increases with the increase in the number of tables;
2. Ensuring that information is distributed to horizontally distributed tables in the database based on a certain rule. During data processing, it helps the distribution module to determine from which table to find the data to be processed and increases the accuracy of the query result;
3. Distribute data as evenly as possible to horizontally distributed tables in the database. Equal distribution of data to tables helps to minimize the time required for operations performed on each table.

Designing the process of distributing data flows in the internal distribution mechanism requires complex algorithms and approaches. These approaches and algorithms are described in the next steps of the study.

**Table 4.1** Analysis of data processing approaches on a single server based on the internal distribution mechanism and a distributed computing system on various indicators.

| Features | Single servers with internal distribution mechanism | Distributed computing systems with multiple servers |
|---|---|---|
| **The price** | + Costs are incurred only for the formation of a single computing machine. Therefore, it is several times cheaper than distributed computing systems that include several computing machines | - Expensive because it includes not only several computers, but also special technical tools that ensure their cooperation |
| **Manageability** | + Managing processes on a single computing machine is always easy. Because the processes in it will not depend on other servers | - It is difficult to manage the system because it is necessary to always take into account other computing machines in the system when managing the organized processes |
| **Security** | + The possibility of setting up security control on a single computing machine is high. Because it is possible to set specific restrictions for entering and exiting the computing system | - There may be a security threat in the communication of each computer to another computer. The process of determining this is complicated |
| **Availability** | - If the calculator is not working for any reason, the system will be unavailable. Ensuring the continuous availability of the system requires additional measures | + The replica of data can be stored on several servers. This allows to hide one server's inactivity through another server |
| **Completeness** | + Data is stored in a single memory. Because of this, it is | - Because the data is repeated in several memories, it is difficult to ensure integrity |

| Features | Single servers with internal distribution mechanism | Distributed computing systems with multiple servers |
|---|---|---|
| | easy to ensure integrity | |
| **Network dependency** | + The system on a single computer does not need to be connected to another computer. Therefore, it does not depend on any network failures | - Since the computers in the system are connected through a communication channel, any network failure will have a negative effect on the system's operation |
| **Scalability** | - No scalability | + The problem of data volume increase can be solved by adding new servers to the system |
| **Parallelism** | ± Parallelism can only be created using internal processors | + Highly efficient parallelism can be achieved through multiple servers |

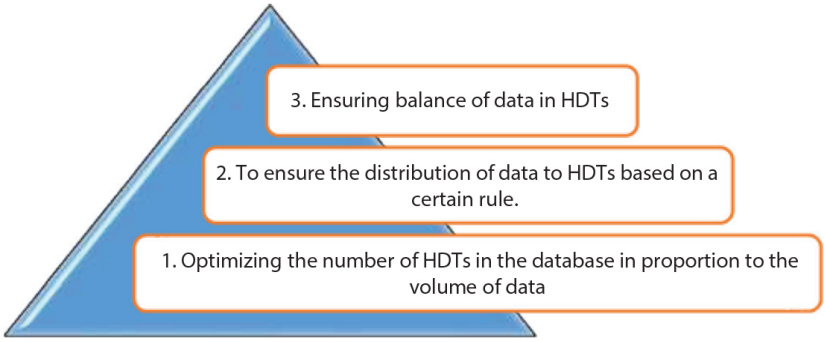



Figure 4.4 Issues that need to be solved for the design of the distribution module in the internal distribution mechanism.

## 4.2 Methodology

### 4.2.1 Method

When automating the management of data flows in systems with an internal distribution mechanism, special attention should be paid to the design rules of the distribution module mentioned above. This process is carried out in a narrow sense in the following steps. In the first step, the determination of the number of horizontally distributed tables proportional to the volume of data is automated. In the second step, the process of intelligent distribution of data is automated.

Instead, in the internal distribution mechanism, the determination of the optimal number of horizontally distributed tables in proportion to the

volume of data is carried out in four stages. In the first stage, the data that can affect the time of queries executed on the data in the horizontally distributed tables is collected based on the results of the experimental test. Also, at this stage, among the collected data, those with a significant correlation with the time indicator are selected [47–49]. This process is carried out on the basis of the functional scheme shown in Figure 4.5.

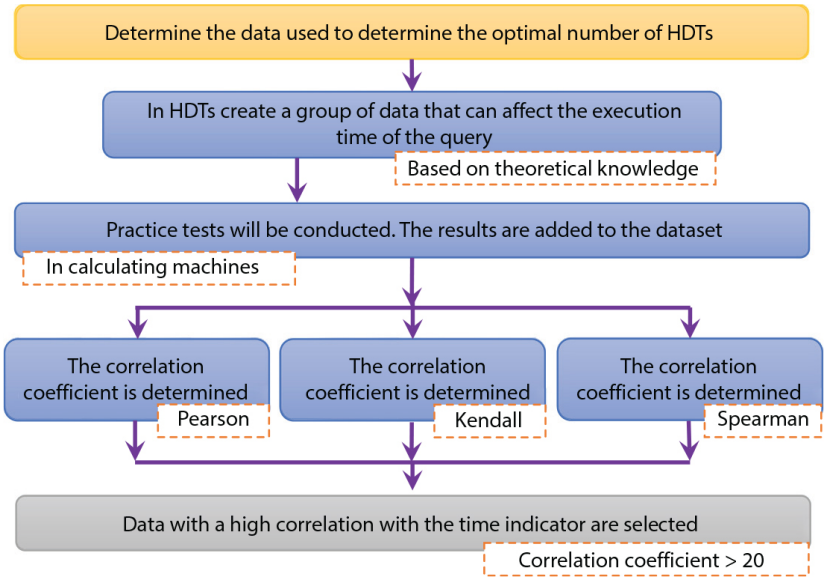



**Figure 4.5** Functional scheme for determining the data group used to determine the optimal number of HDTs.

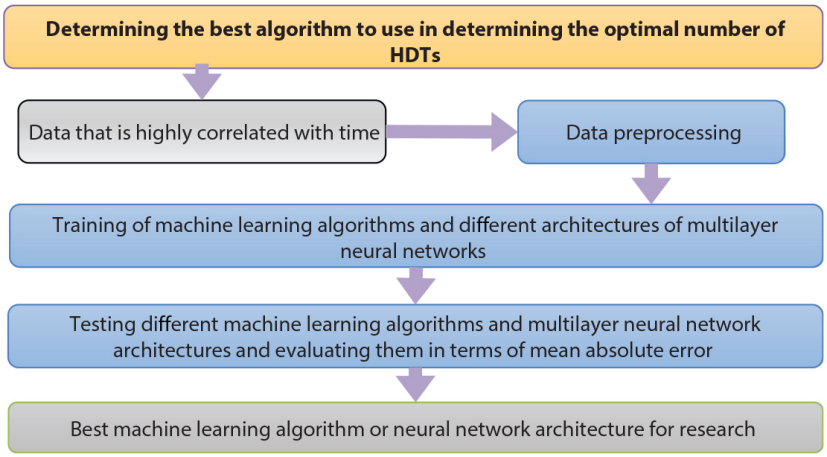



**Figure 4.6** Functional scheme for determining the best algorithm used to find the optimal number of horizontally distributed tables.

In the second stage, the selected group of data is subjected to preliminary processing (scaling, encoding, removing duplicates). In the third step, preprocessed data is sent to machine learning algorithms and multilayer neural networks. Finally, in the fourth stage, it is determined that the neural network and machine learning algorithms have the smallest absolute error compared to the test set. This algorithm is chosen to determine the optimal number of GTJs in the database. These steps are illustrated in the functional diagram in Figure 4.6 above.

The solution of problems 2 and 3, which must be solved for the design of the distribution module in the internal distribution mechanism, is related to the data distribution rule. The distribution rule is found based on the steps in the functional scheme shown in Figure 4.7 below.

Based on the developed distribution rule, data is distributed to GTJs. Based on clustering based on the modulus of vector length presented in the study, the optimal balance of data in GTJs is achieved by changing the limits of vector length in the data distribution algorithm.

In general, the mathematical formulation of the problem of automating the management of data flows in systems with an internal distribution mechanism was expressed as follows:

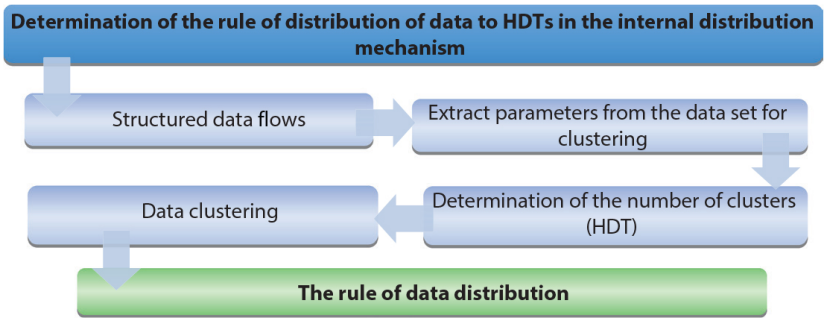



Figure 4.7 Functional scheme for determining the best algorithm used to find the optimal number of horizontally distributed tables.

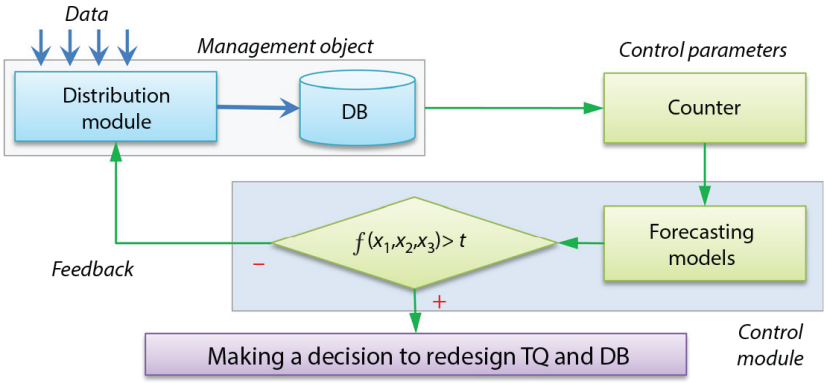



Figure 4.8 Scheme of the model of optimal management of structured data flows.

$$\begin{cases} x_1^{(i)}, x_2^{(i)}, x_3^{(i)} \geq 0; \\ f\left(x_1^{(i)}, x_2^{(i)}, x_3^{(i)}\right) \rightarrow min; \end{cases} \quad (4.8)$$

Here, $x_1$ is the number of horizontally distributed tables, $x_2$ is the total volume of data in horizontally distributed tables, $x_3$ is the total number of records in horizontally distributed tables, $f(x_1,x_2,x_3)$ is the average query corresponding to $x_1$, $x_2$, and $x_3$, and processing time is determined based on machine learning and multilayer neural network predictive models. Based on the above, the scheme of the model of optimal management of structured data flows is represented in Figure 4.8 above.

The model of optimal management of structured data flows is implemented in the following stages:

- Structured data is accepted;
- received data is forwarded to the distribution rule;
- the distribution module saves data to the database;
- the number of horizontally distributed tables in the database, the volume of data, and the number of records in them are determined;
- $f(x_1,x_2,x_3)$ is defined based on forecasting models;

- if the calculated $f(x_1,x_2,x_3)$ is greater than the specified time $t$, the decision to redesign the distribution rule ($TQ$) and the database ($DB$) is made, otherwise, feedback is established in the control system.

In the next sections of the study, the process of developing algorithms for the optimal distribution of structured data streams is studied.

### 4.2.2 Algorithm for Managing Structured Data Streams Based on the Internal Distribution Mechanism

Management of structured data flows based on the internal distribution mechanism can be considered as a combination of two interdependent functions. The first function performs the task of receiving structured data streams, that is, storing them divided into tables with horizontal distribution. The second function performs the task of processing the data stored in these tables, that is, the correct organization of requests. The first function performs complex processes in the management of structured data streams. One of such processes is the process of increasing the number of tables with horizontal distributions in the database in accordance with the volume of stored data. In general, increasing the number of horizontally distributed tables in proportion to the volume of stored data is carried out on the basis of the block-scheme shown in Figure 4.9.

As can be seen from the block scheme in Figure 4.9, the process of managing structured data flows based on the internal distribution mechanism, like all processes, continues until the amount of free memory allocated for the database of the computing machine remains. The limits specified in the first condition of this algorithm are the limits set on the total number of records in the tables for the table to be divided into 2, 3, 4, etc. horizontally distributed tables. The process of determining these limits is a complex process, and the limits are determined not only based on the number of records but also on the execution time of the queries. That is, short request execution time requires smaller threshold values.

Because the limits on the number of records are query-dependent, initial limits require running test queries on specific horizontally distributed tables. In the implementation of this experiment, a table with 40 million rows of records stored in the database of a single computer was used. Based on the information in this table, 5 different experiments are conducted. In the first tests, search queries were performed on the table itself with 40 million rows of records. In the subsequent experiments, the data in this table was first divided into 2, 13, 26, and 49 horizontally distributed tables based on a certain rule. Then, the search queries of the first experiment were performed separately in a group of tables, each with horizontal distributions.

During this experiment, it is required to record the time of queries, the number of horizontally distributed tables (HDTs) at the time of that query, and the total number of records in them. Determining the time of requests is carried out based on the sequence of the algorithm shown in the following block scheme (Figure 4.10).

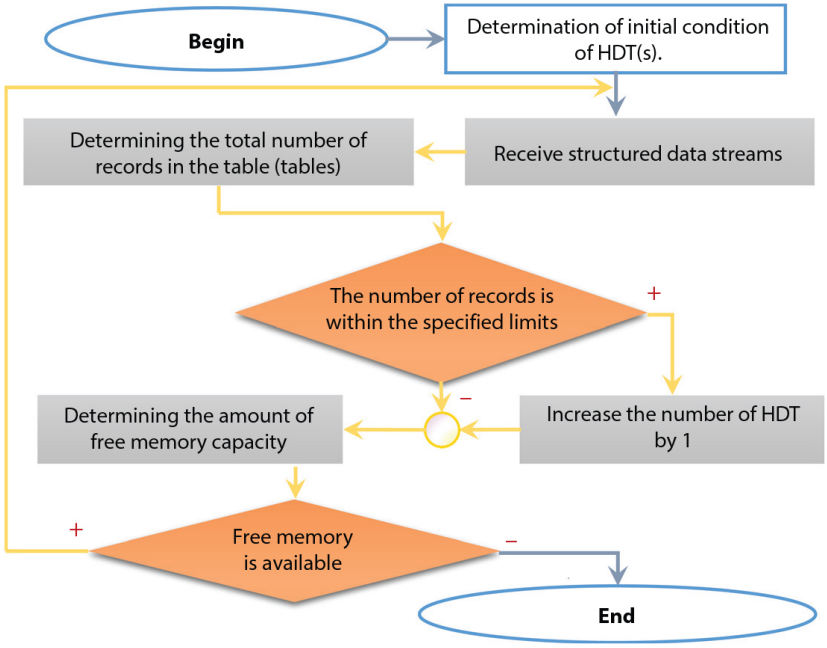



**Figure 4.9** Algorithm for increasing the number of horizontally distributed tables in proportion to the volume of stored data.

As you can see from the algorithm shown in Figure 4.10, when a query is entered, the system instant time is determined, and then the query is passed to the query distribution function. The distribution function forwards the query to one of the current horizontally distributed tables. When the query is completed, the current time of the system is determined again, and the time taken for the query is obtained by subtracting the time of the query from it. The number of horizontally distributed tables queried (n) and the total number of records in them (N) are determined, and the three figures found are stored in memory for use in experiments. The results of the survey conducted on the basis of the algorithm for determining the time of requests in the above HDTs are presented in Table 4.2 below.

Experimental results show that a table with 40 million records does not require table partitioning to process search queries in an average of 23.25 seconds. To process search queries in 7.23 seconds on average, this data should be split into 2 tables with an average of 20 million records each. Similarly, these queries need to be split into 13 tables with an average of 3.079 million records each to process in 1.33 seconds on average, and so on. So, in the data set presented in the sample, the distribution limit of records for processing 7.23 seconds of search queries on the computer used in the experiment is 20 million, for processing 1.33 seconds, 3.079 million, for processing 0.82 seconds, 1.538 million, 0.77 seconds should be set to 0.816 million for processing.

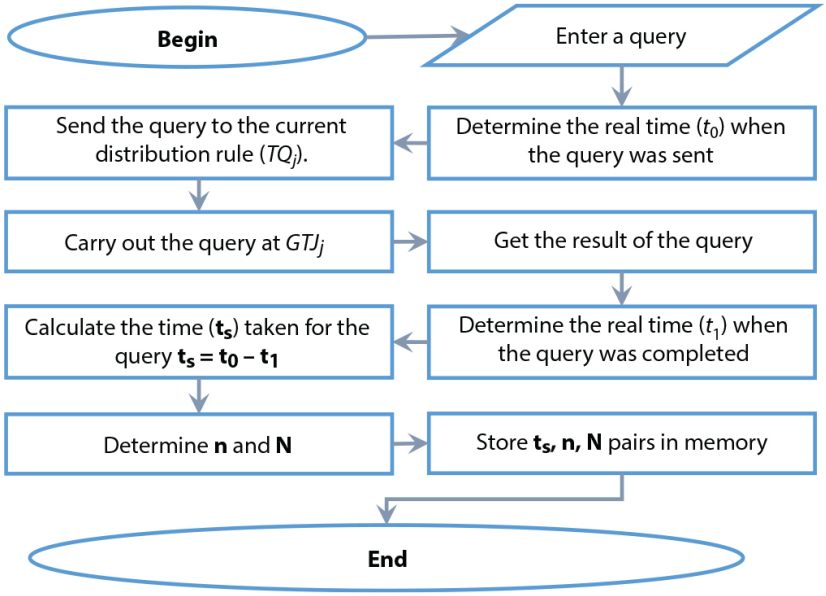



**Figure 4.10** Algorithm for determining the time of queries in horizontally distributed tables.

**Table 4.2** Average time taken for the same search queries on different numbers of horizontally distributed tables (all groups of distributed tables store the same 40 million records).

| **Number of horizontally distributed tables** | **1** | **2** | **13** | **26** | **49** |
|---|---|---|---|---|---|
| **Average time taken for search queries (seconds)** | 23,25 | 7,23 | 1,33 | 0,82 | 0,77 |
| **Average number of entries in GTJs in each group** | 40 mln | 20 mln | 3.079 mln | 1,538 mln | 0,816 mln |

But the above limits are appropriate only for these cases. The formula for finding the query time (4.6) and the obtained experimental results are used to determine the limits for all general cases. In this formula (4.6), $t'_1 \cdot N'$ is the time spent on the distribution process, which is directly proportional to the number of horizontally distributed tables ($n$), so $w_1 \cdot n$ (where $w_1$ is the proportionality coefficient) can be obtained. Query execution time in horizontally distributed tables is inversely proportional to the number of horizontally distributed tables and directly proportional to the total number of records ($N$) in horizontally distributed tables, so it can be obtained in the form of $w_2 \cdot \frac{N}{n}$. In a computing machine, such as the communication time, the time of data passing through the circuits is taken as a constant, and if it is set as $w_0$, the query processing time (t) for the internal distribution mechanism is expressed by the following formula (4.9).

$$t = w_0 + w_1 \cdot n + w_2 \cdot \frac{N}{n} \quad (4.9)$$

As mentioned above, if $\frac{N}{n}$ is taken as the distribution limit ($N_{ch}$) for processing queries at time $t$ in $n$ horizontally distributed tables, it is expressed by the following formula (4.10).

$$N_{ch}(n,t) = \frac{t}{w_2} - \left( \frac{w_0}{w_2} + \frac{w_1}{w_2} \cdot n \right) \quad (4.10)$$

It can be seen from this formula (4.10) that $N_{ch}$ is a two-variable function depending on $t$ and $n$. By putting the $t$ and $n$ obtained in the experimental results into this function, $w_0$, $w_1$, and $w_2$ are found. By putting the values of $w_0$, $w_1$, and $w_2$ found as a result into formula (4.10), the limit of distribution for an arbitrary pair of numbers $t$ and $n$ is determined. For $i$($i$ = 1...$n$) horizontally distributed tables, if the query time t is constant, the boundaries of the distributions ($k$[]) are determined by the following equation:

Automation of Data Flow Management based on Artificial Intelligence in Systems with an Internal Distribution Mechanism 15

$$k[i] = N_{ch}(i,t) \quad (4.11)$$

Taking into account the above, the algorithm for managing data flows in the structured fast data class based on the internal distribution mechanism can be expressed in the following steps:

Step 1. The initial database ($MB_{H0}$) is formed.

Step 2. It is assumed that $i$ = 0 and $N$ = 0 ($i$ represents cases, $N$ is the total number of records in horizontally distributed tables).

Step 3. Initial threshold values are set and written to the $k$[] array.

Step 4. The initial rule ($TQ_i$) for distributing data into horizontally distributed tables is formed.

Step 5. Data is entered.

Step 6. The total number of records ($N$) in horizontally distributed tables is increased.

Step 7. The total number of records ($N$) in horizontally distributed tables is compared to the threshold ($k$[$i$]). If $N$ is less than the limit, go to step 8, otherwise go to step 11.

Step 8. Information is stored in the current database ($MB_{Hi}$) based on the current distribution rule ($TQ_i$) ($Data \rightarrow TQ_i \rightarrow MB_{Hi}$).

Step 9. The amount of free memory ($V$) allocated for the computer database is determined.

Step 10. If there is enough (free) memory, go to step 5, otherwise go to step 18.

Step 11. The value of $i$ is increased by one.

Step 12. The number of horizontally distributed tables is increased and a new state of the database ($MB_{Hi}$) is formed.

Step 13. A new rule ($TQ_i$) for distributing data into horizontally distributed tables is formed.

Step 14. The data in the old database state ($MB_{Hi-1}$) is moved to the new database state ($MB_{Hi}$) based on the new allocation rule ($TQ_i$) ($MB_{Hi-1} \rightarrow TQ_i \rightarrow MB_{Hi}$).

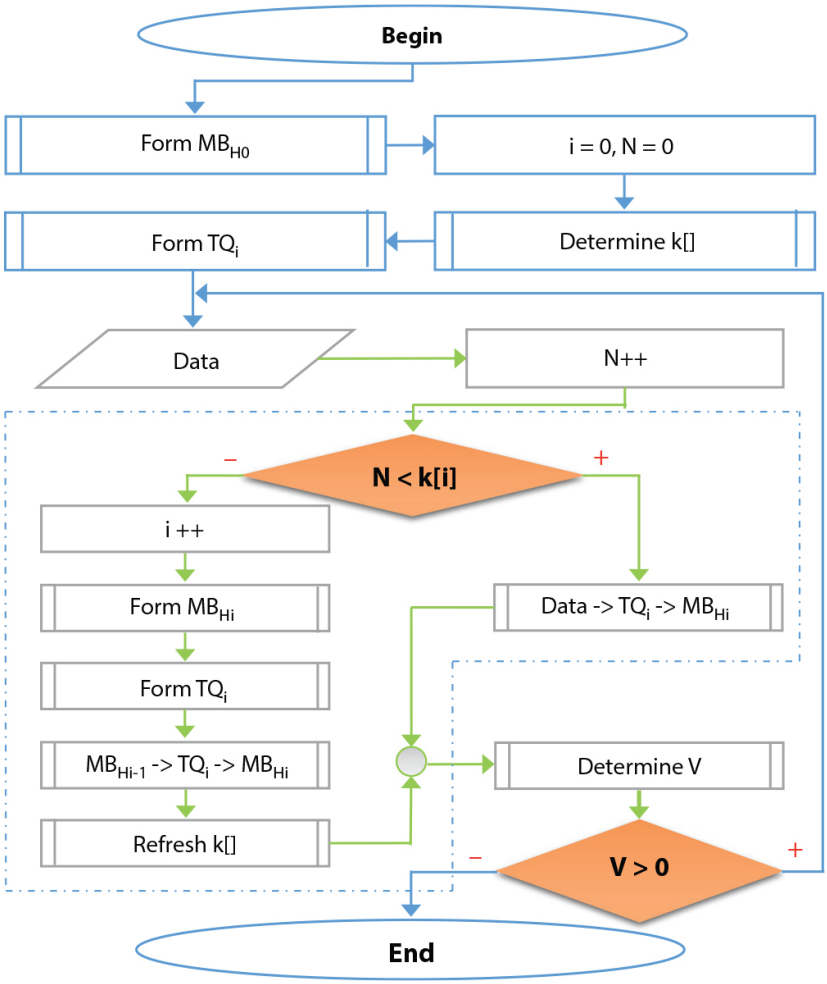



Figure 4.11 Block-scheme of the algorithm for controlling structured data flows based on the mechanism of internal distribution.

Step 15. The limits of records ($k$[]) are updated according to the queries made in the current database in the system.

Step 16. The amount of free memory ($V$) allocated for the computer database is determined.

Step 17. If there is free memory, go to step 5, otherwise go to step 18.

Step 18. The process is completed for this calculator.

The implementation process of this algorithm is represented by the block diagram presented in Figure 4.11.

As a result of the implementation of this algorithm in computing machines, the planned storage of structured data streams is achieved. As a result, the time required for actions performed on these data is reduced. To further improve efficiency, it is necessary to determine the optimal value of the number of horizontally distributed tables and implement the distribution rule in proportion to the optimal number of HDTs. These processes will be covered in the following paragraphs of the study.

### 4.2.3 Optimization of the Number of Internal Distributions Based on Artificial Intelligence

It was noted above that the effectiveness of the internal grafting approach depends on the number of distributions. That is, in order to achieve high efficiency in information systems with an internal distribution mechanism, it is necessary to determine the optimal number of distributions. In this section, methods for finding the optimal number of horizontally distributed tables proportional to the volume of data are considered. To achieve this goal, regression algorithms such as multivariable linear regression, polynomial regression, and logistic regression were considered in the research [50–52]. At the same time, a multilayer neural network was used to determine the number of horizontally distributed tables in proportion to the volume of data. In addition, 6 scaling methods were used in the research to scale the data. These are MinMax, standard, robust, MaxAbs, quantile transformer, and power transformer scaling methods [53–57].

During the study, the five-layer neural network architecture used to determine the optimal number of distributions when the data volume changes on a single computer showed the highest accuracy. In the input layer of this neural network architecture, 3 types of data are input. These are the total volume of data in horizontally distributed tables (Volume), the total number of records (Records), and the time it takes to process these data (Figure 4.12). There are 3 neurons in the second layer, the first hidden layer, and 6 neurons in the second and third hidden layers. As an activating function, “logistic” - logistic, i.e., the sigmoid function was taken. “lbfgs” - Brayden-Fletcher-Golfarb-shanno algorithm was used to optimize the weighting coefficients [58–61].

In order to choose the optimal number of epochs when building the most efficient architecture of a multilayer neural network, the number of epochs was increased in the range [1:500] and the root mean square error (RMSE) was calculated depending on the number of neurons (Figure 4.13). It can be seen from Figure 4.13 that the best result was recorded when epochs were equal to 100.

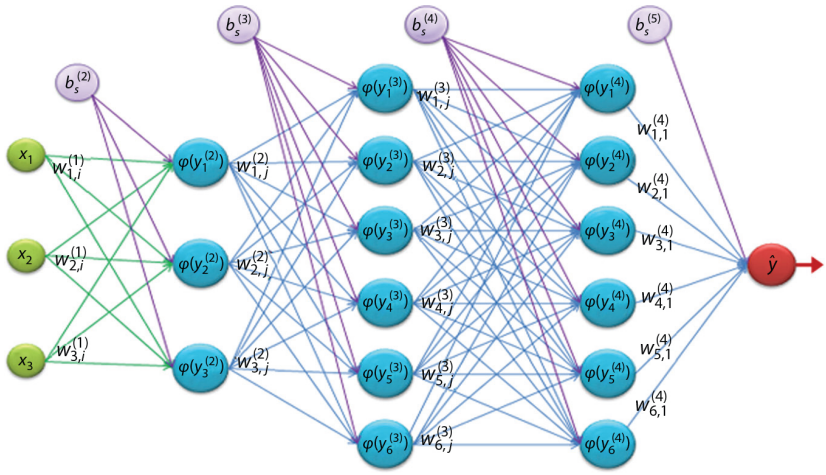


Figure 4.12 Five-layer neural network architecture used in the study.

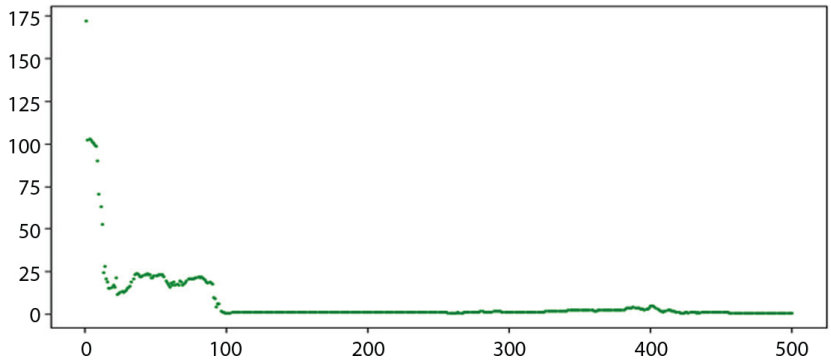



Figure 4.13 Dependence of RMSE on the number of epochs in a neural network.

### 4.2.4 Algorithms for Distribution of Data Streams in the Internal Distribution Mechanism

As mentioned in the above sections, another factor that is important for achieving the intended efficiency from the application of the internal distribution mechanism in information systems is the optimal distribution of data. It is almost impossible to carry out this process under human supervision due to the complexity of the pre-estimation of data flows. Therefore, it is necessary to automate this process with the help of artificial intelligence algorithms. Another reason this process is important is to remember which data is stored in which horizontally distributed table. Remembering the table in which the data is stored is a necessary tool for the subsequent stages of the system, i.e., for quickly searching for data and processing it. If it is not clear in the system which information is stored in which table, this reduces the reliability of the system in the process of operation and has a negative effect on the time efficiency of the system.

The internal distribution mechanism uses the basic approach of fast data storage and processing to determine which table the data is stored in. That is, the data is subjected to a preliminary analysis at the time of entering the system, not after it is saved to the system, and based on the results of this analysis, it is determined in which table it will be saved. This process is carried out based on the data distribution rule ($TQ_i$) corresponding to the optimal number of distributed horizontal tables. Data-sharing rules change as the volume of data increases. The rules used in the system to distribute data into horizontally distributed tables are

called current rules and are also used for queries that are performed on the stored data. That is, what information is stored in which tables is determined based on current rules.

During this study, hierarchical, k-means, bisecting k-means, and unsupervised clustering algorithms were considered in order to develop rules for distributing data into horizontally distributed tables [45, 62, 63]. When clustering data using these algorithms, the length of the records in the selected fields of the incoming structured data and the values associated with the Unicode codes of the characters of these records (the Unicode code value of the first character of the record, the sum of the Unicode codes of all the characters in the record) as parameters received. Here, the set of parameter values of each data can be viewed as the coordinate of this data in three-dimensional Euclidean space. That is, information $A$ can be considered as a point $A(x_1, x_2, x_3)$ in the three-dimensional Euclidean space.

Algorithms for the distribution of data flows were developed based on hierarchical, k-means, bisecting k-means, and birch algorithms, which are unsupervised clustering algorithms. The developed distribution algorithms based on hierarchical, k-means, bisecting k-means, and birch clustering methods are evaluated according to the possibility of equal distribution [64–66]. The results of the study showed that the data distribution algorithms of these data distribution algorithms did not perform well. Therefore, in the research, an algorithm for distributing data flows using clustering based on the square of the vector length is developed in order to distribute the data into horizontally distributed tables as evenly as possible.

The proposed clustering algorithm based on the square of the vector length is implemented by performing the following steps [67]:

Step 1. Let the elements of data set $A$ be entered.

Step 2. The specific characteristics of each information should be transferred to vector coordinates in digital form.

Step 3. Calculate the square of the modules of the resulting vectors.

Step 4. Let the squares of the modules of the vectors be sorted in ascending order.

Step 5. Let the squares of the modules of the vectors sorted in ascending order be written in the array $A'$.

Step 6. Let $n$ be the number of clusters.

Step 7. Let the array $A'$ be divided into $n$ equal parts.

Step 8. Determine the boundaries of the sections.

Step 9. Find the boundaries of the cluster through the boundaries of the segments.

Step 10. Cluster the data through defined boundaries.

Step 11. That is it.

In the algorithm of distribution of data into horizontally distributed tables using clustering based on the square of the vector length, the data that must be permanently stored in memory are the boundaries of the clusters. That is, if the number of clusters is $n$ and the number of cluster boundaries is $n - 1$, the memory complexity of this algorithm is $O(n - 1)$.

The time complexity of the algorithm for dividing data into horizontally distributed tables using clustering based on the square of vector length is $O(m \cdot (\ln(m)+1))$ when considering the initial data into clusters. Here $O(m \cdot (\ln(m)+1))$ represents the time complexity of clustering data with m records into $n$ clusters, while $O(n)$ represents the time complexity of distributing new incoming data into one of $n$ clusters.

## 4.3 Results

### 4.3.1 Algorithms for Distribution of Data Streams in the Internal Distribution Mechanism

In the research, multi-linear regression, polynomial regression, and logistic regression were used in the process of finding the optimal number of horizontally distributed tables in proportion to the volume of data. In addition, multilayer neural network architectures were also used. A total of 518 data were used in the study, 90% of which were selected for training data and 10% for test data. This data was subjected to several scaling methods as mentioned in the previous chapters before being sent to the AI algorithms. Since the results of using different scaling methods were different, all cases were written into the results. The experimental results were evaluated by mean absolute error (MAE) and root mean square error (RMSE).

First, the multi-linear regression algorithm was tested in experiments. The results of the experiment are presented in Table 4.3.

It can be seen from Table 4.3 that for this study, the multi-linear regressor algorithm performed best when applied with the Quantile transformer scaling method. However, the average absolute error of 3.7212 is not satisfactory for research.

**Table 4.3** Error rates of the multi-linear regressor algorithm.

| Assessment criteria | Data scaling methods | | | | | |
|---|---|---|---|---|---|---|
| | MinMax | Standard | Robust | MaxAbs | Quantile transformer | Power transformer method="box-cox" |
| **MAE** | 6.3874 | 6.5384 | 6.3874 | 6.3874 | 3.7212 | 6.2762 |
| **RMSE** | 7.9207 | 7.9759 | 7.9207 | 7.9207 | 5.3273 | 7.9544 |

The accuracy indicators of the polynomial regression algorithm used to determine the optimal numbers of horizontally distributed tables based on the data obtained in the study with different scaling methods are shown in Table 4.4 below.

It can be seen from Table 4.4 that the best result (MAE = 3.0769) was achieved when Quantile transformer scaling method was used to determine the optimal number of horizontally distributed tables based on the Polynomial regression algorithm. However, even this indicator is not satisfactory for research.

In determining the optimal number of horizontally distributed tables, the accuracy indicators of the logistic regression algorithm when used with different scaling methods are shown in Table 4.5 below.

In general, it can be seen that the accuracy of determining the optimal number of HDTs based on the logistic regression algorithm is better than the accuracy indicators of the polynomial and multivariable linear regression algorithms (see Table 4.5). While the results of logistic regression show a satisfactory result for the study, the fact that this algorithm is suitable for discrete values in a certain interval may have a negative effect on the results of the study. Therefore, in order to increase the accuracy of the research results, multilayer neural network architectures were used in experimental tests. From the experiments, it became clear that the best result was recorded in the five-layer neural network architecture (the five-layer neural network architecture is presented in Figure 4.12). The results of this five-layer neural network architecture are shown in Table 4.6.

Table 4.4 Polynomial regression algorithm error rates.

| Assessment criteria | Data scaling methods | | | | | |
|---|---|---|---|---|---|---|
| | MinMax | Standard | Robust | MaxAbs | Quantile transformer | Power transformer method="box-cox" |
| MAE | 7.3455 | 7.3455 | 7.3455 | 7.3455 | 3.0769 | 4.5272 |
| RMSE | 9.1423 | 9.1423 | 9.1423 | 9.1423 | 5.4772 | 6.9872 |

Table 4.5 Error indicators of the logistic regression algorithm.

| Assessment criteria | Data scaling methods | | | | | |
|---|---|---|---|---|---|---|
| | MinMax | Standard | Robust | MaxAbs | Quantile transformer | Power transformer method="box-cox" |
| MAE | 8.6153 | 4.9230 | 2.1538 | 8.6153 | 2.3846 | 6.9230 |
| RMSE | 13.254 | 9.1315 | 5.1140 | 13.254 | 5.1365 | 10.466 |

Table 4.6 Error indicators of the logistic regression algorithm.

| Assessment criteria | Data scaling methods | | | | | |
|---|---|---|---|---|---|---|
| | MinMax | Standard | Robust | MaxAbs | Quantile transformer | Power transformer method="box-cox" |
| MAE | 2.5333 | 2.3740 | 1.7603 | 2.7122 | 0.6019 | 3.7349 |
| RMSE | 3.9097 | 3.7080 | 3.3752 | 3.9924 | 0.6595 | 5.8468 |

From Table 4.6, it is known that the five-layer neural network recorded the smallest absolute error (0.6019) and the smallest root mean square error (0.6595) in the quantile transformer scaling method. This means that the average error of horizontally distributed tables is less than 1 and is considered satisfactory for research use.

### 4.3.2 Results of Automating the Data Distribution Process Based on Artificial Intelligence Algorithms in the Internal Distribution Mechanism

In the above chapters, it is important to distribute data as evenly as possible to horizontally distributed tables in order to achieve efficiency in the internal distribution mechanism. For this, it was evaluated according to the indicator of equal distribution compared to the methods of production clustering based on the algorithm. The experiment distributed data with 57,829 records into 3 horizontally distributed tables using tested partitioning algorithms. The experiment is represented by external diagrams in Figure 4.14.

It can be seen from the diagram presented in Figure 4.14 that the distribution algorithm based on k-means clustering algorithm has shown a better result in terms of equal distribution performance compared to the distribution algorithms based on existing clustering algorithms. However, it can be seen from these diagrams that the best performance belongs to the clustering algorithm based on the vector length squared (VUKA) proposed in the study (19277, 19275, 19277). From the results of these experimental tests, it can be seen that the use of the distribution algorithm based on the square of the vector length in the distribution of data into horizontally distributed tables provides high efficiency in the process of processing structured data streams.

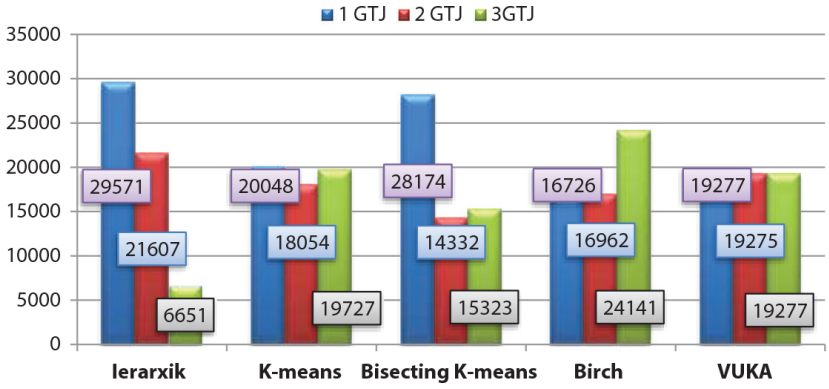



**Figure 4.14** Results of partitioning data with 57829 records into 3 horizontally distributed tables using partitioning algorithms.

## 4.4 Discussion

In this research work, the process of optimal management of data flows in systems with an internal distribution mechanism was considered and a functional scheme of this process was developed. During the research, two main problems of the implementation of the developed functional scheme were identified. These problems are:

- determining the number of distributions (horizontally distributed tables) proportional to the volume of data;
- distribution of data streams into horizontally distributed tables based on clear rules as equal as possible.

Since the first of these problems depends on determining the relationship between the volume of data and the number of distributions, regression

models and multilayer perseptrons-based solutions have been explored. As a result of the research, it was found that the error of solving the first problem using multilayer perseptrons is low. The solution of the second main problem with the help of distribution algorithms based on clustering methods was considered. However, it was found that although the existing clustering methods allow the distribution of data based on a specific rule, they do not always have the possibility of equal distribution. Therefore, a distribution algorithm based on the square of the vector length was developed to distribute the data flows to horizontally distributed tables as evenly as possible and on the basis of a clear rule. By combining these developed algorithms and artificial intelligence models, a control algorithm was developed based on the mechanism of internal distribution of data flows.

It should be noted that the internal distribution mechanism is a representative of distributed computing approaches. However, this approach is fundamentally different from other distributed computing approaches and is a new research direction. Therefore, the literature review section was not included in the study.

In the study, the results obtained from linear and polynomial regressions were not sufficient to determine the number of optimal distributions in proportion to the volume of data. The main reason for this is that there is no correlation between the size of the data and the number of optimal distributions according to these regression lines. Therefore, the multilayer neural network algorithm, which is considered a heuristic method, recorded the best result. Also, the algorithm for distributing data to horizontally distributed tables based on the k-means clustering method recorded a relatively equal distribution indicator in the experimental test. However, because the k-means method is based on the center, it is effective only when the data is evenly distributed around k centers, and in other cases, it does not give the expected result. Therefore, in the study, a distribution algorithm based on the square of the vector length was developed, which allows the distribution of the data as evenly as possible.

Although the internal distribution mechanism is efficient in processing large volumes of data, it may not be efficient in use in information systems serving thousands of clients because the work efficiency in this case depends not only on the system architecture or its algorithms but also on the throughput of the monochannel connecting the clients and the system. That is, although the system works well, the system may not be able to respond to clients requests in time due to monochannel congestion. In the future, in order to solve this problem, research on the use of a combination of a small distributed computing system and an internal distribution mechanism will be considered. In general, the use of an internal distribution mechanism with automated management based on artificial intelligence in storing and processing large amounts of data provides high efficiency not only in terms of time, but also economically.

## 4.5 Conclusion

As stated in the study, the sharp increase in digital data requires the development of both economic and time-efficient methods and algorithms for processing these data. It is known that distributed computing systems are highly efficient in terms of processing and storing large data streams. However, this approach is economically expensive. The internal distribution approach, which is considered to be both economical and time-effective, was considered in the study. However, in this approach, there are such processes that it was determined that the human factor is powerless in managing these processes. In order to overcome this problem in the study, processes of automation of data flow management in systems with internal distribution mechanisms based on artificial intelligence were considered. The results of the research showed that artificial intelligence multilayer neural network architecture and clustering approaches provide high opportunities for managing data flows in systems with an internal distribution mechanism.

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## Note

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